

Activity 2: Errors and Uncertainties in Computations

Table 2.1. Subtractive cancellation in the quadratic formula

Root	Standard formula	Reformulated	Relative error (standard)
x_1 (small root)	-0.000333333370463151	-0.000333333370370379	2.78×10^{-10}
x_2 (large root)	-2999.99966666666293	-2999.99966666666296	1.52×10^{-16}

Table 2.2. Taylor series of $\sin x$ (tolerance 10^{-8})

x	Number of terms	Series result	Relative error
0.1	3	0.099833416667	1.99×10^{-10}
1	6	0.841470984648	1.90×10^{-10}
10	20	-0.544021113721	5.21×10^{-9}
30	48	-0.988278332189	2.50×10^{-4}

VI. Analysis Questions

- Why does the standard quadratic formula fail for one root when $b^2 \gg 4ac$? Show the exact step where digits are lost.

Answer: For the small root, the formula involves subtracting two nearly equal large numbers:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \frac{-3000 + 3000.0000005}{2}.$$

The subtraction cancels most of the significant digits, causing loss of precision (subtractive cancellation).

- For the $\sin x$ series at $x = 30$, the individual terms grow as large as $\sim 10^{12}$ before shrinking. Explain why this happens and why it affects accuracy even though the series converges.

Answer: When x is large, the terms $x^{2k+1}/(2k+1)!$ initially increase in magnitude before eventually decreasing. These huge intermediate values cause round-off and cancellation errors, so even though the series converges, significant digits are lost and the final result becomes inaccurate.

- Why is downward recursion stable where upward recursion is unstable? Relate your answer to the two independent solutions of the recurrence.

Answer: Upward recursion amplifies small round-off errors because the dominant solution grows with l , while the desired solution is very small. Downward recursion suppresses the growing solution and keeps the desired solution stable. The two independent solutions are the growing and decaying solutions of the recurrence.

- In Figure 2.1 the downward-recursion error sits near 10^{-16} . What sets this floor?

Answer: The floor is set by machine precision (double precision floating-point), about 10^{-16} , which is the smallest relative difference that can be represented.

VII. Conclusion

This activity shows that a mathematically correct formula is not necessarily numerically stable. Subtractive cancellation, round-off and truncation errors, and algorithmic instability can all lead to significant inaccuracies. Careful reformulation, argument reduction, and choosing the right algorithm (e.g., downward recursion) are essential for reliable numerical results.

Table 2.1. Subtractive cancellation in the quadratic formula

Root

Standard formula

X1 (small root)

-0.000333333370463151

X2 (large root)

-2999.99966666666293

Table 2.2. Taylor series of $\sin x$ (tolerance 10^{-8})

Reformulated

Relative error (standard)

-0.000333333370370379

2.78×10^{-10}

-2999.99966666666296

1.52×10^{-16}

x

Number of terms

Series result

Relative error

0.1

3

0.099833416667

1.99×10^{-10}

1

6

0.841470984648

1.90×10^{-10}

10

20

-0.544021113721

5.21×10^{-9}

30

48

-0.988278332189

2.50×10^{-4}

VI. Analysis Questions

1. Why does the standard quadratic formula fail for one root when $b^2 \gg 4ac$? Show the exact step where digits are lost.

Answer: For the small root, the formula involves subtracting two nearly equal large numbers:

$$-b \pm \sqrt{b^2 - 4ac}$$

$$x_1 =$$

$$2a$$

$$-3000 + 3000.0000005$$

The subtraction cancels most of the significant digits, causing loss of precision (subtractive cancellation).

2. For the $\sin x$ series at $x = 30$, the individual terms grow as large as $\sim 10^{12}$ before shrinking. Explain why this happens and why it affects accuracy even though the series converges.

Answer: When x is large, the terms $x^{2k+1}/(2k+1)!$ initially increase in magnitude before eventually decreasing. These huge intermediate values cause round-off and cancellation errors, so even though the series converges, significant digits are lost and the final result becomes inaccurate.

3. Why is downward recursion stable where upward recursion is unstable? Relate your answer to the two independent solutions of the recurrence.

Answer: Upward recursion amplifies small round-off errors because the dominant solution grows with 1 , while the desired solution is very small. Downward recursion suppresses the growing solution and keeps the desired solution stable. The two independent solutions are the growing and decaying solutions of the recurrence.

4. In Figure 2.1 the downward-recursion error sits near 10^{-16} . What sets this floor?

Answer: The floor is set by machine precision (double precision floating-point), about 10^{-16} , which is the smallest relative difference that can be represented.

VII. Conclusion

This activity shows that a mathematically correct formula is not necessarily numerically stable. Subtractive cancellation, round-off and truncation errors, and algorithmic instability can all lead to significant inaccuracies. Careful reformulation, argument reduction, and choosing the right algorithm (e.g., downward recursion) are essential for reliable numerical results.